

Mathematical Foundations of Reinforcement Learning and Online Decision Making

Lecture 1 - Markov Chains

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Course Overview

asdf

Markov Chain

A Markov chain is a pair $(\mathcal{X}, P(\cdot))$ of

- 1 **States:** $\mathcal{X}$
- 2 **Transitions:** $P : \mathcal{X} \times \mathcal{X} \rightarrow [0, 1]$ where

$$\sum_{y \in \mathcal{X}} P(x, y) = 1 \quad \text{for all } x \in \mathcal{X}.$$

Markov Chain

A trajectory (realization) of $(\mathcal{X}, P(\cdot))$ is a sequence

$$X_0, X_1, X_2, \dots, X_t, \dots$$

where

$$\begin{aligned} P(X_{t+1}=y|X_t=x, X_{t-1}=x_{t-1}, \dots, X_0=x_0) &= P(X_{t+1}=y|X_t=x) \\ &= P(x, y) \end{aligned}$$

Frog Process

States:

$$\mathcal{X} = \{e, w\}$$

Transitions:

$$P = \begin{pmatrix} P(e, e) & P(e, w) \\ P(w, e) & P(w, w) \end{pmatrix} = \begin{pmatrix} 1-p & p \\ q & 1-q \end{pmatrix}$$



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Question

If we start in state e what is the probability of being in state e after t time steps?

Frog Process

State distributions:

- 1 $\mu_0(x)$ is the probability that $X_0 = x$.
- 2 $\mu_t(x)$ is the probability that $X_t = x$

Question

If we know μ_0 can we compute μ_t ?

$$\mu_t = \mu_{t-1}P = \mu_0 P^t$$



Frog Process

Questions

- What happens to μ_t when t becomes large? (i.e., does the limit $\lim_{t \rightarrow \infty} \mu_t$ exist?)
- If μ_t converges to some value, how fast does it converge?

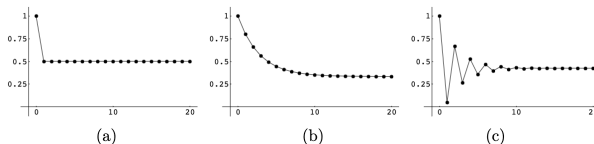


FIGURE 1.2. The probability of being on the east pad (started from the east pad) plotted versus time for (a) $p = q = 1/2$, (b) $p = 0.2$ and $q = 0.1$, (c) $p = 0.95$ and $q = 0.7$. The long-term limiting probabilities are $1/2$, $1/3$, and $14/33 \approx 0.42$, respectively.

Frog Process

General Rule

A system on the form

$$x_{t+1} = f(x_t)$$

can only converge to one of its fixed points, i.e., to x^* such that $x^* = f(x^*)$. **But the system is not guaranteed to converge.**

Frog Process

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Let

$$\pi = \lim_{t \rightarrow \infty} \mu_t$$

(if exists), then we must have

$$\pi = \pi P$$

Frog Process

By solving the linear system $\pi = \pi P$ we get

$$\pi(e) = \frac{q}{p+q} \quad \pi(w) = \frac{p}{p+q}$$

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Then we get the linear scalar system

$$\begin{aligned} \Delta_{t+1} &= \mu_t(e)(1-p) + (1-\mu_t(e))(q) - \frac{q}{p+q} = (1-p-q)\Delta_t \\ &= (1-p-q)^t \Delta_0 \end{aligned}$$

Frog Process

$$\Delta_{t+1} = (1 - p - q)^t \Delta_0$$

converges to zero at an exponential rate if $p < 1$ and $q < 1$. So

$$\lim_{t \rightarrow \infty} \mu_t(e) = \frac{q}{p + q} \quad \text{and} \quad \lim_{t \rightarrow \infty} \mu_t(w) = \frac{p}{p + q}.$$

Frog Process

Can we say about the convergence speed?

Frog Process

Can we say about the convergence speed?

Yes,

$$|\mu_t(e) - \pi(e)| \leq |1 - (p + q)|^t \Delta_0$$

So we can $|\mu_t(e) - \pi(e)| \leq \epsilon$ for all

$$t \geq \frac{1}{p + q} \log \left(\frac{\Delta_0}{\epsilon} \right).$$

Frog Process

What can we say more generally?

- Do all Markov chains have stationary distributions?

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What can we say more generally?

- Do all Markov chains have stationary distributions?

Answer: Irreducibility and Aperiodicity

- If there is stationary distribution, how fast do we converge to it?

Answer: Mixing time.

Irreducibility

Irreducible Markov Chain

A Markov chain is called irreducible if for any two states $x, y \in \mathcal{X}$ there exists integer t such that $P^t(x, y) > 0$.

- Can think of irreducibility as a connectivity of the underlying graph”
- Irreducibility is also a property of the Markov Matrix P (that is where the name comes from).

Aperiodicity

Period of a state

Define

$$\mathcal{T}(x) = \{t \geq 1 : P^t(x, x) > 0\}.$$

Then the period of a state is defined as the greatest common divisor of $\mathcal{T}(x)$.

Lemma

In an irreducible Markov Chain all states have the same period. This common period is called the period of the irreducible Markov chain.

Aperiodicity

Irreducible and Aperiodic Markov Chain

If an irreducible Markov Chain has the period 1 then it is Irreducible and Aperiodic.

Lemma

In an irreducible Markov Chain all states have the same period. This common period is called the period of the irreducible Markov chain.

Examples



FIGURE 1.3. Random walk on $\mathbb{Z}_{10}$ is periodic, since every step goes from an even state to an odd state, or vice-versa. Random walk on $\mathbb{Z}_9$ is aperiodic.

Discussion

- Can we force a process to be aperiodic?

Stationary Distribution

Lemma

We call a distribution π on $\mathcal{X}$ a stationary distribution if

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- **Question 1:** Do we converge to a stationary distribution for any initialization?

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Lemma

We call a distribution π on $\mathcal{X}$ a stationary distribution if

$$\pi = \pi P$$

- **Question 1:** Do we converge to a stationary distribution for any initialization?
- **Question 2:** Does a stationary distribution even exist?

Question 1

If a Markov Chain has a unique stationary distribution, can we converge converge to that stationary distribution under any (or some) initializations?

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Generally no

Consider

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

and $\pi = [0.5, 0.5]$.

Question 1

Theorem

If a Markov chain is irreducible and aperiodic and has an stationary distribution π then

$$\lim_{t \rightarrow \infty} \mu_t = \pi$$

for all distributions on the initial states.

Question 2

Hitting Time

We define the hitting time for x as

$$\tau_x = \min\{t \geq 0 : X_t = x\}.$$

We also consider also

$$\tau_x^+ = \min\{t > 0 : X_t = x\}$$

and when $X_0 = x$ we call τ_x^+ the first hitting time.

Question 2

Theorem

A irreducible Markov chain has a unique stationary distribution $\pi : \mathcal{X} \rightarrow [0, 1]$ where

$$\pi(x) = \frac{1}{\mathbb{E}_x(\tau_x^+)}$$

Mixing Time

Questions

- How long does it take a markov chain to converge to its stationary distribution?
- How long time does it take for a Markov chain to forget" its initial state/distribution?

Mixing Time

Total Variation

Total variation distance between two distributions μ and π

$$\|\mu - \pi\|_{TV} = \max_{A \subseteq \mathcal{X}} |\mu(A) - \pi(A)|$$

or equivalently

$$\|\mu - \pi\|_{TV} = \frac{1}{2} \sum_{x \in \mathcal{X}} |\mu(x) - \pi(x)|$$

Mixing Time

Theorem

If $(\mathcal{X}, P)$ is a irreducible and aperiodic Markov chain with stationary distribution π then there exists $\alpha \in (0, 1)$ and C such that

$$\max_{x \in \mathcal{X}} \|P^t(x, \cdot) - \pi\| \leq C\alpha^t$$

$$d(t) = \max_{x \in \mathcal{X}} \|P^t(x, \cdot) - \pi\|$$

Mixing Time

$$d(t) = \max_{x \in \mathcal{X}} \|P^t(x, \cdot) - \pi\|$$

Mixing Time

The mixing time is defined as

$$t_{\text{mix}}(\epsilon) = \min\{t : d(t) \leq \epsilon\}$$

Mixing Time - Examples

- The mixing time for our Frog process is
$$t_{mix}(\epsilon) = \frac{1}{p+q} \log \left(\frac{\epsilon}{\Delta_0} \right)$$
- Lazy random walk on a cycle is $t_{mix} = \Theta(n^2)$
- Random walk on Hypercube $t_{mix} = \mathcal{O}(n \log(n))$

Mixing Time - How to compute it

Mostly Upper bounds:

- Coupling (Chapter 5)
- Strong Stationary Times (Chapter 6)
- Relationship with cover times (Chapter 11)
- Eigenvalues of the transition matrix P (Chapter 12)

Lower bounds are discussed in Chapter 7

Gamler's Ruin

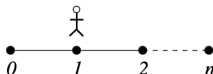


FIGURE 2.1. How long until the walk reaches either 0 or n ? What is the probability of each?

Gambler moves up by 1 and down by 1 by equal probability. The process stops when she either reaches state 0 or n .

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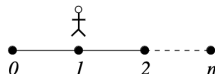


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Questions

Given that the gambler starts in state k

- what is the probability that she reaches n ?
- what is the expected time until completion? (i.e., until she reaches 0 or n)

Gamler's Ruin

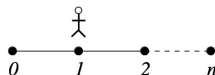


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Theorem

Let X_t be the gamler's fortune at time t and let τ be the time until final state 0 or n . Assume that $X_0 = k$ where $0 \leq k \leq n$. Then

$$P(X_\tau) = \frac{k}{n}$$

$$\mathbb{E}(\tau) = k(n - k)$$

Gamler's Ruin

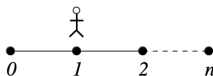


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Proof solve simple linear system.

Coupon Collecting

A company issues n different types of coupons. A collector desires a complete set. We suppose each coupon he acquires is equally likely to be each of the n types. How many coupons must he obtain so that his collection contains all n types?

It may not be obvious why this is a Markov chain. Let X_t denote the number of different types represented among the collector's first t coupons. Clearly $X_0 = 0$. When the collector has coupons of k different types, there are $n - k$ types missing. Of the n possibilities for his next coupon, only $n - k$ will expand his collection. Hence

$$\mathbf{P}\{X_{t+1} = k + 1 \mid X_t = k\} = \frac{n - k}{n}$$

and

$$\mathbf{P}\{X_{t+1} = k \mid X_t = k\} = \frac{k}{n}.$$

Every trajectory of this chain is non-decreasing. Once the chain arrives at state n (corresponding to a complete collection), it is absorbed there. We are interested in the number of steps required to reach the absorbing state.

Coupon Collecting

PROPOSITION 2.3. *Consider a collector attempting to collect a complete set of coupons. Assume that each new coupon is chosen uniformly and independently from the set of n possible types, and let τ be the (random) number of coupons collected when the set first contains every type. Then*

$$\mathbf{E}(\tau) = n \sum_{k=1}^n \frac{1}{k}.$$